Supplementary Appendix for Enforcement and Bargaining over Illicit Drug Wholesale Prices: Structural Evidence from a Transnational Asian Gang

A.1 Additional Figures and Tables

	Ice	Ecstasy	Ketamine	Erimin
Drug Class	A	A	B	C
Maximum prison years	Life	20	20	10
Maximum strokes of the cane	15	15	10	5
Threshold presumed trafficking*	25g	10g		
Mandatory Death Penalty	250g			

*Presumed trafficking may have a maximum sentence of life in prison.

Table A.1: Drug-trafficking laws in Singapore during our sample period.

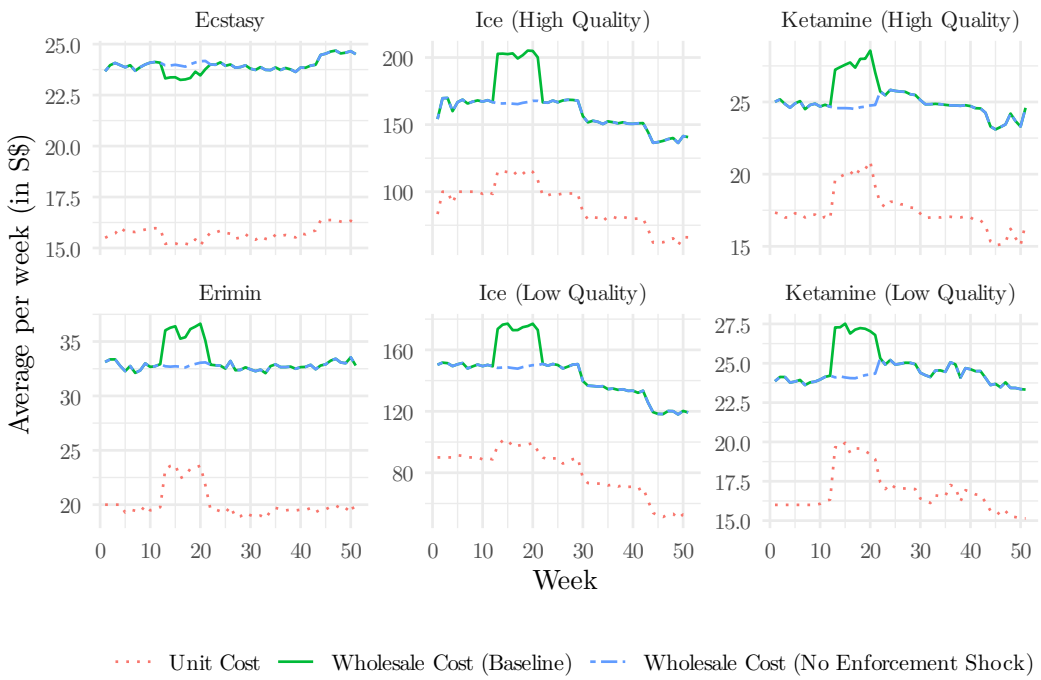


FIGURE A.1: AVERAGE WEEKLY WHOLESALE PRICES, NO ENFORCEMENT SHOCK VERSUS BASELINE.

A.2 End-User Prices

Ecstasy

The United Nations Office for Drug Control and Crime Prevention report that the average price of one dose of ecstasy in Singapore in 1998 was US$26.[1] Using the average closing exchange rate during our year of data, this is approximately S$43. This is the end-user price per tablet that we use in estimation. Interviews with ex-drug offenders reveal that the prices varied between S$35-60 per tablet, so S$43 falls within the range. Furthermore, 99% of trades in our data had a wholesale price of S$30 or less, so at S$43 pushers still make considerable gross margins.

[1] https://www.unodc.org/pdf/report_1999-06-01_1.pdf

Erimin

Obtaining accurate price data for erimin is more difficult. Ex-drug offenders report that the price per slab (10 pills) of erimin costed between S$60-100 during our sample period. We take the midpoint of this range and use S$80 per slab, or S$8 per pill. Although several years since our sample period, a 2009 UNODC report gives a price of US$4 per pill in 2006 (S$6.36).[2] A separate newspaper article reports that prices ranged from S$6-8 per pill in 2006.[3] 99% of trades in our data have a wholesale price of S$60 or less per slab, so at S$80 pushers make large gross margins.

Ice

The United Nations Office for Drug Control and Crime Prevention report that the average price of one gram of ice in Singapore in 1999 was US$147.1, or S$249.[4] End users report a price range of S$150-380 per gram. A newspaper article reports that the price was between S$120-180 for half a gram.[5] The 25th and 75th percentiles of wholesale prices in the data are S$130 and S$180 per gram. In the data, the wholesale price for high-quality ice is on average 13.5% larger than for low-quality ice. Therefore we use S$240 for the end-user price for low-quality ice and S$280 for high-quality ice, which falls within the ranges given by the interviews and the reports, and has a similar difference in price to the average difference in wholesale prices for high- and low-quality ice.

Ketamine

Obtaining accurate price data for ketamine is also difficult. Ex-drug offenders report a range of S$45-85 per gram for the end-user price. An article from 2006 states that the price ranged from S$40-50.[6] The wholesale prices for

[2] http://drugslibrary.wordpress.stir.ac.uk/files/2017/03/2009_Patterns_and_Trends.pdf
[3] https://www.thestar.com.my/news/regional/2006/02/15/addicts-turn-to-erimin5
[4] https://www.unodc.org/pdf/report_2002-06-26_1/report_2002-06-26_1.pdf
[5] https://www.asiaone.com/health/rising-number-ice-users-singapore
[6] https://www.thestar.com.my/news/regional/2006/02/15/addicts-turn-to-erimin5

high- versus low-quality ketamine are quite similar. The medians differ by only 4%. Therefore we use S$45 and S$50 per gram for low- and high-quality ketamine respectively.

A.3 Obtaining Nash Bargaining Wholesale Prices

To calculate the wholesale prices $\boldsymbol{w}_{it}$ that satisfy the Nash bargaining first-order conditions for products where $p_{jt} > c_{jt} + \xi_{ijt}$, we find the fixed point of the function $\boldsymbol{\Omega}(\boldsymbol{w}_{it})$, whose jth element is:

$$\Omega_j(\boldsymbol{w}_{it}) = p_{jt} - \xi_{ijt} - \frac{(1-\beta_i)(p_{jt} - \xi_{ijt} + c_{jt} - 2w_{ijt})\, v_{it}(\boldsymbol{w}_{it})}{\beta_i \pi_{it}(\boldsymbol{w}_{it})}$$

We use the bisection method to find the fixed point of this function. The algorithm is as follows:

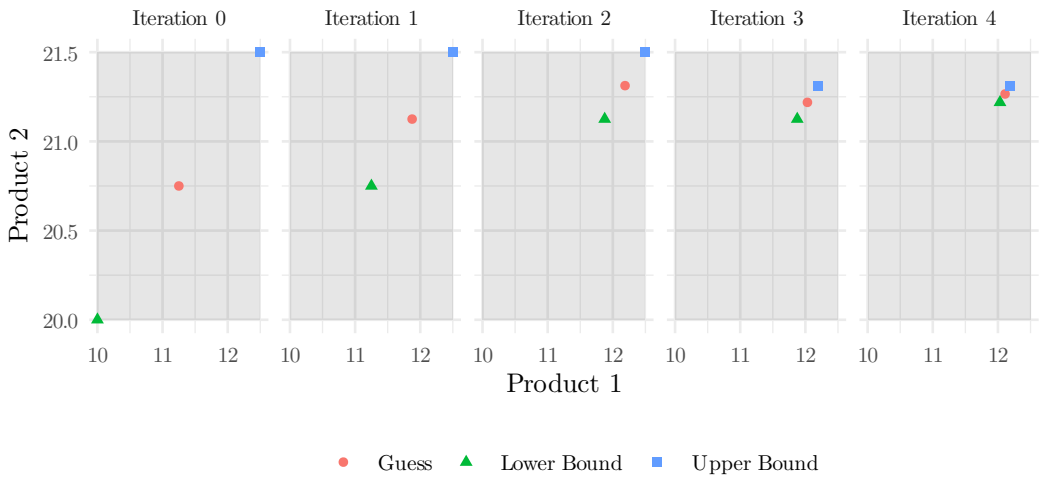
Step 1 Set initial ($k = 0$) lower and upper bounds and initial guess:

- Set the initial lower bound to $\underline{w}^0_{ijt} = c_{jt} + \delta(p_{jt} - \xi_{ijt} - c_{jt})$, where δ is a small constant (such as $10 \times e^{-5}$) such that $\Omega_j(\underline{\boldsymbol{w}}^0_{it}) < \underline{w}^0_{ijt}$ for all $i, j, t,$. The small constant is to avoid dividing by zero when $\pi_{it}(\boldsymbol{w}_{it}) = 0$.
- Set the initial upper bound to $\overline{w}^0_{ijt} = \frac{p_{jt} - \xi_{ijt} + c_{jt}}{2}$, such that $\Omega_j\left(\overline{\boldsymbol{w}}^0_{it}\right) > \overline{w}^0_{ijt}$ for all $i, j, t,$.
- Set the initial guess $w^0_{ijt} = \frac{1}{2}\left(\underline{w}^0_{ijt} + \overline{w}^0_{ijt}\right)$.

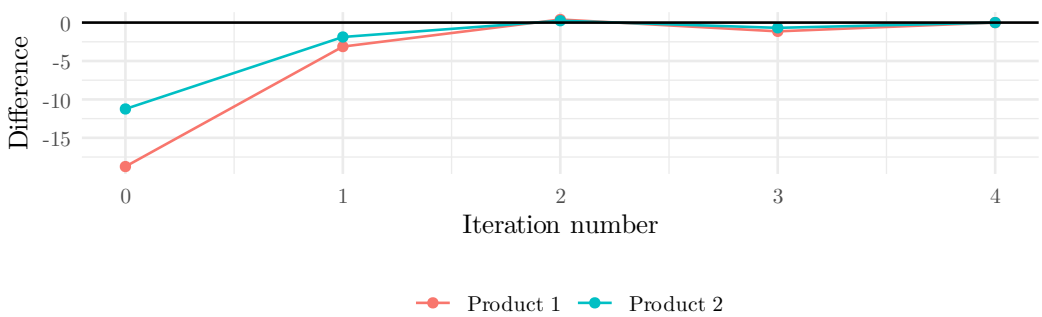
Step 2 For iteration $k > 0$, update the bounds and guess:

- If $\Omega_j\left(\boldsymbol{w}^k_{ijt}\right) < w^k_{ijt}$, set $\underline{w}^{k+1}_{ijt} = w^k_{ijt}$. Otherwise, set $\underline{w}^{k+1}_{ijt} = \underline{w}^k_{ijt}$.
- If $\Omega_j\left(\boldsymbol{w}^k_{ijt}\right) > w^k_{ijt}$, set $\overline{w}^{k+1}_{ijt} = w^k_{ijt}$. Otherwise, set $\overline{w}^{k+1}_{ijt} = \overline{w}^k_{ijt}$.
- Set $w^{k+1}_{ijt} = \frac{1}{2}\left(\underline{w}^{k+1}_{ijt} + \overline{w}^{k+1}_{ijt}\right)$.

Step 3 Repeat step 2 incrementing k until $\|\boldsymbol{\Omega}(\boldsymbol{w}_{it}) - \boldsymbol{w}_{it}\|$ is below a pre-specified tolerance level.



(A) Iterating wholesale prices.



(B) Difference in $\Omega_j\left(\boldsymbol{w}_{it}^k\right) - w_{ijt}^k$ for $j = 1, 2$.

Figure A.2: Fixed point iteration example with two products.

An example with two products is shown in Figure A.2. This would occur in practice if there were two products where $p_{jt} > c_{jt} + \xi_{ijt}$ and $p_{jt} \leq c_{jt} + \xi_{ijt}$ for the remaining four products. The sum of marginal costs are $(c_{1t} + \xi_{i1t}, c_{2t} + \xi_{i2t}) = (10, 20)$ and the prices are $(p_{1t}, p_{2t}) = (15, 23)$. The initial lower bound $\underline{\boldsymbol{w}}_{it}^0$ is the green triangle and the initial upper bound $\overline{\boldsymbol{w}}_{it}^0$ is the blue square in Figure A.2a. The initial guess $\boldsymbol{w}_{it}^0$ is the red circle at the midpoint of these two points. In Figure A.2b, the difference $\Omega_j\left(\boldsymbol{w}_{it}^0\right) - w_{ijt}^0$ is negative for both products. Therefore the lower bound for iteration 1 is updated to $\underline{w}_{it}^1 = \boldsymbol{w}_{it}^0$ for iteration 1 and the upper bound $\overline{\boldsymbol{w}}_{it}^1$ remain the same.

The green triangle for iteration 1 moves to the position of the red circle from iteration 0. At $\boldsymbol{w}_{it}^1$, the difference $\Omega_j\left(\boldsymbol{w}_{it}^1\right) - w_{ijt}^1$ is again negative for both products, so only the lower bound is updated. In iteration 2, $\Omega_j\left(\boldsymbol{w}_{it}^2\right) - w_{ijt}^2$ is positive for both products, so $\overline{\boldsymbol{w}}_{it}^3$ is set equal to $\boldsymbol{w}_{it}^2$ and the blue square in iteration 3 moves to the position of the red circle in iteration 2. This process continues until convergence. In practice, prices converge after 20-30 iterations.

A.4 Legalization Counterfactual

In this section we consider an additional counterfactual experiment where ice is legalized. We focus on ice because the gang we study was the only gang in the country selling ice at the time, while there were several other gangs actively selling the other drugs. For ice, we observe the total quantity sold on the end-user market and can use the enforcement shock to estimate the slope of the end-user demand curve. While no country has legalized ice to date, there has been a discussion of decriminalizing the drug in Australia (McGowan, 2019). Due to data limitations, this counterfactual requires a number of additional assumptions, but we will argue that we estimate a lower bound on the tax revenue that could be raised, and that the actual tax revenue is likely to be much larger.

We assume that the gang becomes a centralized importer and the pushers become licensed wholesalers. This is similar to the case of prostitution in Singapore. When prostitution was legalized, a fixed number of permits were issued to individuals already active in the market (see, for example, Li et al. (2018)). We assume the centralized importer's marginal costs are the same as the gang's, and since the enforcement shock would not occur for ice under legalization, we use the same marginal costs as described in the no enforcement shock counterfactual. We assume each pusher's demand is unchanged, but since the penalties from being caught are removed we argue that pushers would actually sell more ice under legalization. We further assume that the end-user demand function is unchanged following legalization. However, given the results in Jacobi and Sovinsky (2016), we would expect end-user demand to

increase following legalization as access would increase and the social stigma of drug consumption would be reduced. We consider a tax rate of 7%, which is equal to the sales tax rate in Singapore and is in the lower end of the state taxes on marijuana in the US and Canada.[7]

In our data, we observe the total equilibrium quantity during the no-shock period and the enforcement shock period. We also observe the equilibrium price during the no-shock period and have estimated the rationalized price change during the enforcement shock period with our model. Since the enforcement shock is a cost shock which shifts the supply curve, we observe two points on the end-user market demand curve. If we make a linear approximation of the end-user demand curve, the enforcement shock allows us to estimate the slopes of the end-user demand curve for high- and low-quality ice. This approximation enables us to predict end-user demand after the imposition of a tax.[8] We also highlight that in our model each individual pusher faces a perfectly elastic demand curve since the end-user market is competitive. However, the market as a whole may face a less elastic demand curve. We find the demand elasticities for high- and low-quality ice to be -0.46 and -0.50, respectively.

To calculate how the burden of the tax is shared between end users and pushers, we use the following procedure. Given a trial share of tax burden, we calculate total end-user demand using our linear approximation and total pusher supply by simulating trades according to our estimated bargaining model. We then find the vector of new equilibrium end-user prices such that total demand equals total supply in each product.

We find that end users bear 72.2% of the tax on high-quality ice and 69.8% of the tax on low-quality ice. This is in line with low-quality ice being slightly more elastic than high quality. The tax reduces the total quantity sold on the market by 2.67% and 2.83% for high- and low-quality ice, respectively. For some intuition about how the pushers' relative bargaining power with the gang

[7] In the US, excise taxes on marijuana vary between 7% and 37% and in Canada between 3.9% and 19.3%.

[8] To calculate these slopes, we only use weeks 10-48 in the data, where the number of active pushers is close to the median value.

affects the tax incidence between pushers and end users, we also calculate the incidence shares if we shift all the pushers' bargaining parameters to just above zero and just below one. We find that if pushers have more bargaining power over the gang, that the end users bear even more of the tax. The opposite occurs if the pushers have less bargaining power.

The estimated lower bound on the annualized tax revenue is S$738,621 (approximately US$760,000 in 2019 dollars), where 53.4% of this is from high-quality ice. Singapore's population was just under 4 million during our sample period. Therefore the tax intake from legalization may be very small and it is likely that the accompanying administrative costs would outweigh it.

References

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